

Abstract

This page reviews Linear Algebra, from the perspective of linear functionals. We deem that readers already know the basic facts and main results of linear algebra: unique representation of vectors with respect to a basis, determinant, eigenvectors and eigenspaces, direct sum and projection, inner product and the spectral decomposition.

0.1 Preliminaries

Recall that: the philosophy of basis.

Lemma 0.1.1. Let V be a Vector Space over the field F . A subset $\beta \subset V$ is a basis for V if and only if: For any vector $x \in V$, there exist unique finite non-zero scalars $a_i \in F$ and distinct vectors $v_i \in \beta$ such that

$$x = a_1v_1 + a_2v_2 + \cdots + a_nv_n$$

The 'unique' implies that:

if $x = a_1v_1 + \cdots + a_nv_n = b_1w_1 + \cdots + b_mw_m$ where $a_i, b_i \in F \setminus \{0\}$ and $v_i, w_j \in \beta$ are distinct within each sum, then $n = m$ and for some permutation σ , $v_i = w_{\sigma(i)}$, $a_i = b_{\sigma(i)}$ for all $1 \leq i \leq n$.

Since the representation is unique, we can define the *coordinate representation* with respect to the basis β .

Definition 0.1.2. Let $\beta = \{v_1, \dots, v_n\}$ be an ordered basis for V . The *coordinate representation* (or *coordinate map*) with respect to β is defined as:

$$[\cdot]_{\beta} : V \rightarrow F^n : x \mapsto \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

where $x = \sum_{i=1}^n a_i v_i$.

Definition 0.1.3. Let $W_1, \dots, W_k$ be subspaces of a vector space V .

We say that V is the *direct sum* of $W_1, \dots, W_k$, denoted by $V = \bigoplus_{i=1}^k W_i$, if:

1. $V = \sum_{i=1}^k W_i = \{w_1 + \cdots + w_k \mid w_i \in W_i\}$
2. For each j , the intersection of W_j and the sum of the other subspaces is trivial:

$$W_j \cap \left(\sum_{i \neq j} W_i \right) = \{0\}$$

Definition 0.1.4. Let $V = W_1 \oplus W_2$. Define a *projection* onto W_1 along W_2 as:

$$p_1 : V \rightarrow V : x_1 + x_2 \mapsto x_1$$

where $x_1 \in W_1$, $x_2 \in W_2$. It is well-defined due to the unique representation of vectors in a direct sum.

Proposition 0.1.5. Let $V = W_1 \oplus W_2 \oplus \cdots \oplus W_k$, and p_j be the projection onto W_j along the subspace $\bigoplus_{i \neq j} W_i$. These operators satisfy the following properties:

1. $p_j^2 = p_j$ for all j .
2. $p_i p_j = 0$ for $i \neq j$.
3. $\sum_{i=1}^k p_i = I$, where I is the identity operator on V .
4. $\text{Im}(p_j) = W_j$ and $\text{Ker}(p_j) = \bigoplus_{i \neq j} W_i$.

0.2 Dual Space

Let V be a vector space over a field F .

Definition 0.2.1. A linear map $f : V \rightarrow F$ is called a *linear functional* on V .

Example. Let $F[x]$ be the set of polynomials over F . For each $\alpha \in F$, define

$$L_\alpha : F[x] \rightarrow F : p(x) \mapsto p(\alpha)$$

This is called the 'evaluation' at α . Clearly, it is a linear functional since

$$L_\alpha(c \cdot p(x) + q(x)) = c \cdot p(\alpha) + q(\alpha) = c \cdot L_\alpha(p(x)) + L_\alpha(q(x))$$

Example. Let $C([a, b])$ be the space of continuous real-valued functions on $[a, b]$. Define

$$L : C([a, b]) \rightarrow \mathbb{R} : f \mapsto \int_a^b f(t) dt$$

L is clearly a linear functional by the linearity of the integral.

Definition 0.2.2. The collection of all linear functionals on V , denoted by $\mathcal{L}(V, F)$, is called the *dual space* of V and is denoted as V^* .

Theorem 0.2.3. Let V be a finite-dimensional vector space over F , and let $\beta = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V . Then, there exists a unique *dual basis* $\beta^* = \{f_1, \dots, f_n\}$ for V^* such that $f_i(\alpha_j) = \delta_{ij}$. Furthermore, for each linear functional $f \in V^*$,

$$f = \sum_{i=1}^n f(\alpha_i) f_i$$

and for each vector $\alpha \in V$,

$$\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$$

Proof. By the property of basis, for any $\alpha \in V$, there exist unique scalars $a_1, \dots, a_n \in F$ such that

$$\alpha = a_1 \alpha_1 + \dots + a_n \alpha_n$$

For each $1 \leq i \leq n$, define $f_i : V \rightarrow F$ by $f_i(\alpha) = a_i$ (or $f_i(\alpha) = ([\alpha]_\beta)_i$).

Then $f_i(\alpha_j) = \delta_{ij}$. Since $\dim V = \dim V^*$, it is enough to show that the set $\beta^* = \{f_1, \dots, f_n\}$ is linearly independent. Suppose that for some $c_i \in F$, $\sum_{i=1}^n c_i f_i = 0$. Then for each $1 \leq j \leq n$,

$$\left(\sum_{i=1}^n c_i f_i \right) (\alpha_j) = \sum_{i=1}^n c_i f_i(\alpha_j) = c_j = 0$$

Thus, β^* is linearly independent and forms a basis for V^* .

The second assertion can be proved as follows: if $\alpha \in V$ is represented as $\alpha = \sum_{i=1}^n x_i \alpha_i$ where $x_i \in F$, then for each $1 \leq j \leq n$:

$$f_j(\alpha) = f_j \left(\sum_{i=1}^n x_i \alpha_i \right) = \sum_{i=1}^n x_i f_j(\alpha_i) = x_j$$

Substituting x_j back into the representation of α , we obtain $\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$. □

Application: Lagrange Interpolation

Let $P_n(F) = \{p \in F[x] : \deg p \leq n\}$, and let $t_0, t_1, \dots, t_n \in F$ be distinct elements. For each $0 \leq i \leq n$, define the linear functional $L_i : P_n(F) \rightarrow F$ by $L_i(p) = p(t_i)$. These $\{L_0, L_1, \dots, L_n\}$ are linearly independent. Suppose that

$$c_0 L_0 + c_1 L_1 + \dots + c_n L_n = 0 \quad \dots (*)$$

where $c_i \in F$. By substituting the standard basis $\{1, x, x^2, \dots, x^n\}$ into $(*)$, we obtain a system of equations:

$$\begin{cases} c_0 + c_1 + \dots + c_n = 0 \\ c_0 t_0 + c_1 t_1 + \dots + c_n t_n = 0 \\ \vdots \\ c_0 t_0^n + c_1 t_1^n + \dots + c_n t_n^n = 0 \end{cases} \implies \begin{bmatrix} 1 & 1 & \dots & 1 \\ t_0 & t_1 & \dots & t_n \\ \vdots & \vdots & \ddots & \vdots \\ t_0^n & t_1^n & \dots & t_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The matrix above is the **Vandermonde Matrix**, which is invertible since t_i are distinct. Thus $c_0 = c_1 = \dots = c_n = 0$. More precisely, define a linear map

$$T : P_n(F) \rightarrow F^{n+1} : p(x) \mapsto (p(t_0), \dots, p(t_n))$$

For $\beta = \{1, x, \dots, x^n\}$, $[T]_\beta$ is the **Vandermonde Matrix**.

If we choose a different basis $\beta' = \{1, (x - t_0), (x - t_0)(x - t_1), \dots, \prod_{j=0}^{n-1} (x - t_j)\}$, then $[T]_{\beta'}$ becomes:

$$\det[T]_{\beta'} = \det \begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & t_1 - t_0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_n - t_0 & \dots & \prod_{j=0}^{n-1} (t_n - t_j) \end{bmatrix} = \prod_{1 \leq i < j \leq n} (t_j - t_i) \neq 0$$

Since the determinant is invariant with respects to the similar matrix, it is proved.

Theorem 0.2.4. $\{L_0, \dots, L_n\}$ is a dual basis for $P_n(F)^*$ with respect to the basis $\{P_0(x), \dots, P_n(x)\}$ for $P_n(F)$, where

$$P_i(x) = \prod_{j \neq i} \frac{x - t_j}{t_i - t_j}$$

Since $L_j(P_i) = P_i(t_j) = \delta_{ij}$, for any $p(x) \in P_n(F)$, we have:

$$p(x) = \sum_{i=0}^n L_i(p) P_i(x) = \sum_{i=0}^n p(t_i) P_i(x)$$

Furthermore, by the linearity of the integral, $\int_a^b p(x) dx = \sum_{i=0}^n p(t_i) \int_a^b P_i(x) dx$.

0.3 Annihilators

Definition 0.3.1. Let S be a subset of a vector space V . The **annihilator** of S , denoted by S^0 , is defined as:

$$S^0 = \{f \in V^* : f(\alpha) = 0 \text{ for all } \alpha \in S\}$$

Example. In $P_2(F)$ with basis $\beta = \{1, x, x^2\}$ and dual basis $\{f_0, f_1, f_2\}$ (where f_i extracts the coefficient of x^i), consider $S = \{x\}$. Then $f \in S^0$ if $f(x) = 0$. Since $f = c_0f_0 + c_1f_1 + c_2f_2$, $f(x) = c_1 = 0$. Thus, $S^0 = \text{span}\{f_0, f_2\}$.

Theorem 0.3.2. Let V be a finite-dimensional vector space and W be a subspace of V . Then:

$$\dim W + \dim W^0 = \dim V$$

Proof. Let $\{\alpha_1, \dots, \alpha_k\}$ be a basis for W . Extend it to a basis $\{\alpha_1, \dots, \alpha_n\}$ for V . Let $\{f_1, \dots, f_n\}$ be the dual basis for V^* . For $i \geq k+1$, $f_i(\alpha_j) = 0$ for all $1 \leq j \leq k$, so $f_i \in W^0$. The set $\{f_{k+1}, \dots, f_n\}$ is linearly independent. For any $f \in W^0$, we can write $f = \sum_{i=1}^n f(\alpha_i)f_i$. Since $f \in W^0$, $f(\alpha_i) = 0$ for $i \leq k$. Thus $f = \sum_{i=k+1}^n f(\alpha_i)f_i$, which means $\{f_{k+1}, \dots, f_n\}$ spans W^0 . Therefore, $\dim W^0 = n - k = \dim V - \dim W$. □

0.4 Double Dual

Question: Is every basis for V^ a dual of some basis for V ?*

Lemma 0.4.1. Let V be a finite dimensional vector space over F . For each $\alpha \in V$, define

$$L_\alpha : V^* \rightarrow F : f \mapsto f(\alpha)$$

Then, it is linear functional on V^* . Moreover, if $L_\alpha(f) = 0$ for any $f \in V^*$, then $\alpha = 0$.

Note: the last assertion can be said: $(V^*)^0 = \{0\}$.

Proof. Let $f, g \in V^*$ and $c \in F$ be given. Then,

$$L_\alpha(cf + g) = (cf + g)(\alpha) = cf(\alpha) + g(\alpha) = cL_\alpha(f) + L_\alpha(g)$$

Thus proved. To show the second assertion, let $\alpha \neq 0$. Then, there exists a basis β for V containing α . Then, there is a dual basis β^* for V^* . Take $f_\alpha \in \beta^*$ corresponding to α . Then, $f_\alpha(\alpha) = 1 \neq 0$. Thus proved. $\square$

Theorem 0.4.2. Let V be a finite dimensional vector space over F . For each $\alpha \in V$, define

$$L_\alpha : V^* \rightarrow F : f \mapsto f(\alpha)$$

Then, the map $V \rightarrow V^{**} : \alpha \mapsto L_\alpha$ is an Isomorphism.

Proof. Let $\alpha, \beta \in V$ and $c \in F$, and for any $f \in V^*$,

$$L_{c\alpha + \beta}(f) = f(c\alpha + \beta) = cf(\alpha) + f(\beta) = cL_\alpha(f) + L_\beta(f) = (cL_\alpha + L_\beta)(f)$$

Moreover, by the Lemma, the kernel of the map is trivial. Thus injective, and surjective because:

$$\dim V^{**} = \dim V^* = \dim V$$

$\square$

Corollary 0.4.3. Let V be a finite dimensional vector space over F .

IF $L \in V^{**}$, then there exists a unique $\alpha \in V$ such that for all $f \in V^*$, $L(f) = f(\alpha)$.

Corollary 0.4.4. Let V be a finite dimensional vector space over F .

Every basis for V^* is the dual of some basis for V .

Proof. Let $\{f_1, \dots, f_n\}$ be a basis for V^* . Then, there is a dual basis $\{L_1, \dots, L_n\}$ for V^{**} such that $L_i(f_j) = \delta_{ij}$. By the above corollary, for each L_i , there exists a unique $\alpha_i \in V$ such that $L_i(f) = f(\alpha_i)$ for all $f \in V^*$. Now, $L_i(f_j) = f_j(\alpha_i) = \delta_{ji}$, the claim is proved. $\square$

0.5 Transpose

Theorem 0.5.1. Let V and W be finite dimensional vector space over F , and let $T : V \rightarrow W$ be a linear map. Then,

$$T^t : W^* \rightarrow V^* : g \mapsto g \circ T$$

is a linear map. Furthermore, for any ordered basis β and γ for V and W , respectively,

$$([T]_{\beta}^{\gamma})^t = [T^t]_{\gamma^*}^{\beta^*}$$